

## Solution Requirements

|                      |                                                                  |
|----------------------|------------------------------------------------------------------|
| <b>Date</b>          | 08-07-2026                                                       |
| <b>Team ID</b>       |                                                                  |
| <b>Project Name</b>  | Automated Credit Card Approval Prediction using Machine Learning |
| <b>Maximum Marks</b> | 4 Marks                                                          |

### Define Functional and Non-Functional Requirements:

A comprehensive list of requirements defining exactly what the solution does and how well it performs. •

Functional Requirements (FR) define specific behaviors, functions, or features of the system (What the system should do).

• Non-Functional Requirements (NFR) specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should be).

### Step 1: Functional Requirements (FR)

| S.No | Requirement Category | Requirement Description                                                                                                                                                                                          | Priority (High/Medium/Low) |
|------|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| 1    | Input Collection     | The system shall allow applicants to enter personal and financial details such as Age, Income, Employment Status, Credit History, Loan Amount, Existing Debt, and other required information through a web form. | High                       |
| 2    | Crop Prediction      | The system shall process the submitted applicant details using the trained Machine Learning model and predict whether the credit card application is <b>Approved</b> or <b>Rejected</b> .                        | High                       |
| 3    | Input Validation     | The system shall validate all mandatory fields before prediction and display an error message if any data is missing or invalid.                                                                                 | Medium                     |
| 4    | Result Display       | The system shall display the credit card approval or rejection result immediately after prediction.                                                                                                              | High                       |
| 5    | Navigation           | The system shall provide Home, About, Credit Card Prediction, and Contact pages through a common navigation menu.                                                                                                | Medium                     |
| 6    | Public Access        | The system shall be accessible through a public URL without requiring software installation.                                                                                                                     | High                       |

## Step 2: Non-Functional Requirements (NFR)

| S.No | NFR Category               | Requirement Description                                                                                            | Target Metric / Acceptance Criteria                      |
|------|----------------------------|--------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|
| 1    | Performance & Speed        | The system shall return the prediction result within <b>2 seconds</b> after form submission.                       | Response time < 2 sec                                    |
| 2    | Usability                  | The interface shall be simple and user-friendly so that applicants can use the system without guidance.            | Task completed without external help                     |
| 3    | Reliability & Availability | The application shall remain available during demonstrations and handle invalid or missing input without crashing. | No crashes on invalid/missing input                      |
| 4    | Portability                | The application shall work consistently across major web browsers and systems with Python installed.               | Works on Chrome, Firefox, Edge                           |
| 5    | Maintainability            | The code shall be modular by separating the ML model, preprocessing, prediction logic, and UI components.          | Modular file structure (app.py, templates/, model files) |
| 6    | Accuracy                   | The trained Machine Learning model shall achieve high prediction accuracy for credit card approval decisions.      | ≥ 95% test accuracy (achieved: 97.3%)                    |